Project Design Phase

Problem–Solution

Date	15 Febuary 2025
Team ID	GACSE-114
Project Name	FitBuddy: AI Fitness Plan Generator Using Gemini Model
Maximum Marks	2 Marks

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	• ndividuals often struggle to maintain healthy lifestyles because they lack personalized fitness guidance. • Users face significant confusion due to excessive and conflicting information available online. • Professional trainers and nutritionists come with high costs that are not accessible to everyone. • Existing resources typically provide generic plans that ignore specific body metrics, individual goals, or dietary preferences. • This lack of personalization leads to ineffective routines, low user motivation, and poor overall health outcomes.

2.	Idea / Solution description	• FitBuddy AI is an AI-powered fitness and nutrition assistant designed to generate personalized workout and diet plans. • The system uses user-specific data such as age, weight, height, activity level, diet preference, and fitness goals to tailor its output. • The application calculates essential health metrics, including Body Mass Index (BMI) and Total Daily Energy Expenditure (TDEE), to estimate accurate calorie requirements. • It provides a customized diet plan, structured workout routines, and motivational advice to keep users engaged. • An integrated AI chatbot allows users to ask fitness-related questions and receive real-time guidance.
3.	Novelty / Uniqueness	• FitBuddy AI integrates advanced artificial intelligence with personalized fitness planning to deliver highly tailored health recommendations. • Unlike traditional apps with static, generic plans, this system uses AI to generate dynamic strategies for every unique user. • The platform features interactive AI chat support and a 30-day weight projection chart to help users visualize their fitness journey.
4.	Social Impact / Customer Satisfaction	• The platform promotes healthier lifestyles by providing accessible and affordable fitness guidance to a broad audience. • It increases public awareness about nutrition and helps individuals develop sustainable, long-term fitness habits. • By simplifying the health planning process, it enhances user satisfaction and encourages consistent engagement with healthy routines.

5.	Business Model (Revenue Model)	• The platform utilizes a **freemium model**: basic features like BMI calculations and simple workout suggestions are free. • Advanced features, including AI-generated personalized plans, detailed nutrition tracking, and premium coaching, are available via subscription. • Secondary revenue streams include partnerships with fitness brands, targeted advertisements, and premium health analytics.
6.	Scalability of the Solution	• Designed as a web-based application, the system can easily scale to support a massive global user base. • The solution leverages cloud-based infrastructure and AI APIs to expand services without requiring massive changes to physical infrastructure. • Future growth plans include mobile application development, wearable device integration (e.g., smartwatches), and advanced AI-driven health analytics.